We now focus on applying the steps of the scientific method to estimate and test our market impact model.

SOLUTION TECHNIQUE

The Question

How much will my order cost me to trade?

In other words, how much premium is required to attract additional sellers so I can complete my order; how much do I need to discount my price to attract additional buyers so I can complete my order?

Research the Problem

The research step for our market impact modeling problem consists of both academic research (e.g., literature review), and observation and analysis of actual data.

First, let us start with the academic literature. Madhavan (2000, 2002), highly regarded as the ultimate starting point, provides a detailed review of relevant transaction cost analysis research and the market microstructure literature leading up to algorithmic trading. Almgren and Chriss (1997); Kissell and Glantz (2003); Kissell, Glantz and Malamut (2004); Wagner (1991); Gatheral (2010, 2012); and Domowitz and Yegerman (2006, 2011) provide us with a strong foundation and starting point for algorithmic trading findings.

Our review of the literature provided many key findings. Cost is dependent upon the number of shares traded (e.g., trade size, total order size, or imbalance). This has been shown by Loebb (1983); Holtausen, Leftwich and Mayers (1987); Chan and Lakonishok (1993); Plexus Group (2000) and others. Empirical evidence reveals that costs vary by volatility and market capitalization. For example, see Stoll (1978); Amidhud and Mendelson (1980); Madhavan and Sofianos (1998); Chan and Lakoniskhok (1995); Keim and Madhavan (1997); and Breen, Hodrick and Korajczyk (2002) to name a few.

Price impact results directly from trade and/or information leakage, as well as the liquidity needs of the investor or institutional fund. Market conditions over the trading period highly affect the underlying costs as well. See Beebower and Priest (1980); Wagner and Edwards (1993); Perold and Sirri (1993); Copeland and Galai (1983); and Stoll (1995). Additionally, there has been numerous evidence presented by Kyle (1985), Bertismas and Lo (1998); Grinold and Kahn (1999); and Almgren and